Effect of Hyperparameters on Signature-Based Generators

November 14, 2025

1 Hyperparameters

Let $\{S_t\}_{t=0}^{T^*}$ be a univariate time series. A signature-based generator conditions on the past window of length L to produce H new steps per rollout until a total horizon of T steps is reached. The main hyperparameters are:

- **lookback** (L): Number of past steps used to compute signature features that condition the generator. This parameter applies to signature based methods only.
- **forward** (H): Number of steps produced per rollout. This differs from the total length T; we roll out in multiple shorter segments rather than all at once. In our generators, only piecewise bootstrap method has this parameter.
- **future steps** (T): Total number of steps to generate. With rollouts, the model typically executes $\lceil T/H \rceil$ iterations.
- **signature level** (m): Truncation level of the (log-)signature. Higher levels capture richer higher-order interactions in the history but increase feature dimensionality and computational cost.

Dimensionality note. For input dimension d (after any augmentation), the size of the degree-$\leq m$ tensor signature grows roughly like

$$\sum_{k=1}^{m} d^k = \frac{d(d^m - 1)}{d - 1} \qquad (d > 1),$$

illustrating the rapid increase in feature size as either d or m grows.

2 Evaluation Method

In this experiment, we evaluate generators on known stochastic processes using the truncated-signature MMD computed on time-index–augmented series with a rolling window of 5.

As in real markets, we observe a historical observable path simulated from ground truth process. With known stochastic processes, we can then simulate N future paths. Meanwhile, fitting the historical path, the generator can generate N future paths. If the generated future path is more similar to simulated future path, the generator is better. This similarity is captured by signature MMD.

3 Simulators and Generators

The known stochastic processes include the following with specified parameters.

- geometric Brownian motion (GBM): $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, with $\mu = 0.10$ and $\sigma = 0.30$.
- constant–elasticity–of–variance (CEV) model: $dS_t = \mu S_t\,dt + \sigma S_t^{\beta}\,dW_t$, with $\mu = 0.20$, $\sigma = 0.30$ and $\beta = 0.8$; paths are generated by a Milstein scheme with truncation at zero to avoid negativity.
- Merton jump–diffusion: $\frac{dS_t}{S_{t-}} = (\mu - \lambda\kappa)\,dt + \sigma\,dW_t + (J-1)\,dN_t$, with $\mu = 0.10$, $\sigma = 0.30$, jump intensity $\lambda = 0.5$, log–jump mean $m_J = -0.1$, and log–jump s.d. $s_J = 0.2$; the drift is adjusted to preserve the desired martingale property.
- Variance–Gamma (VG) process (with parameters $\theta = -0.1$, $\sigma = 0.20$, $\nu = 0.10$) under a martingale correction and carrying rates $r = 0.03$, $q = 0$, $\mu = 0.10$, $\sigma = 0.20$.

Unless noted otherwise, all simulations use a business-day step $\Delta t = 1/252$, initial level $S_0 = 100$, a fixed seed (42) for reproducibility.

Generators we considered here include

- piece-wise bootstrap with time augmentation transformation or lead-lag transformation
- hybrid method with time augmentation transformation or lead-lag transformation
- KRR with time augmentation transformation or lead-lag transformation
- ARIMA model
- GARCH model

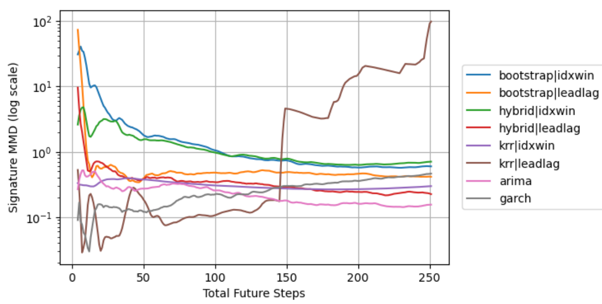
4 Effect on Evaluation

4.1 Total Future Steps (T)

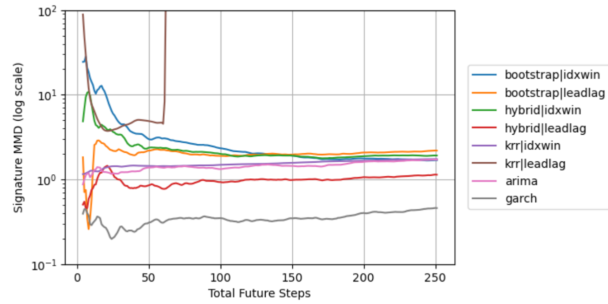
Figure 1 shows the how evaluation results are affected by the total future steps considered. The lookback window length is fixed at 60. Generally, the performance of generators stables as number of total future steps increases. However, KRR method with lead-lag transformation on time series produces unstable and explosive paths.

4.2 Lookback Window (L)

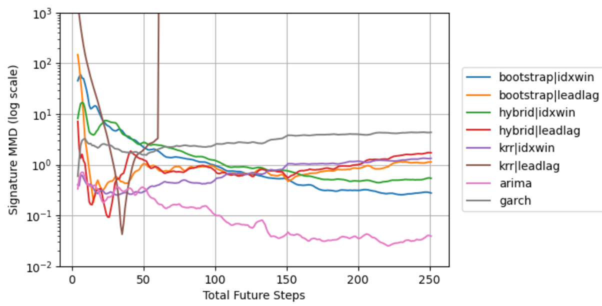
Figure 2 shows how the lookback-window length affects the performance of signature-based generators. The total number of future steps is fixed at 252. The KRR generator with the lead–lag transformation tends to stabilize as the lookback window increases. By contrast, for the bootstrap and hybrid methods, longer lookback windows appear to amplify instability.



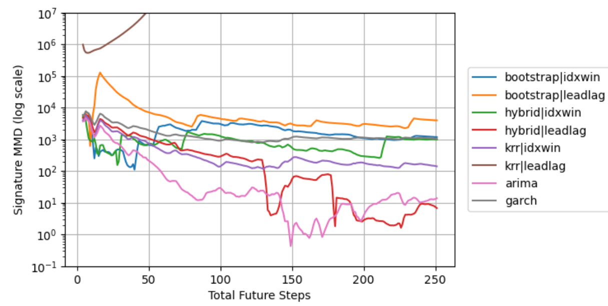
(a) GBM



(b) CEV

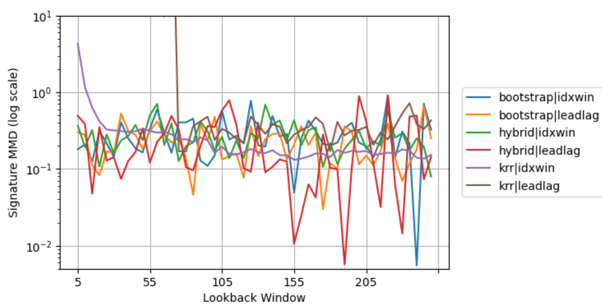


(c) Merton Jump Diffusion

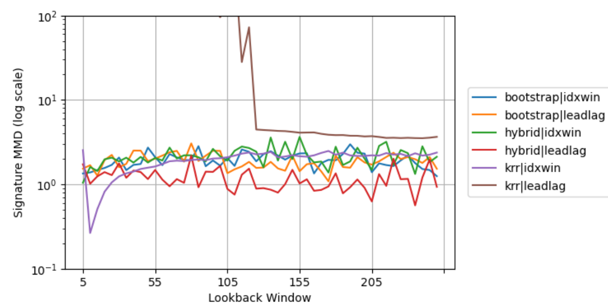


(d) Variance Gamma

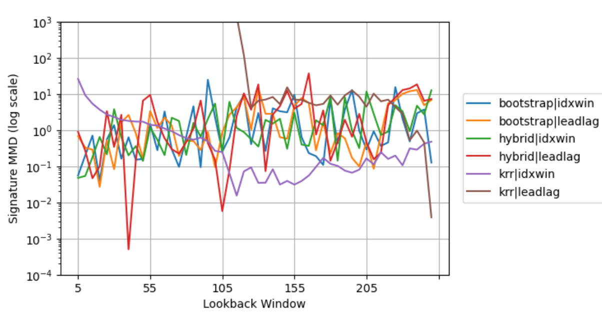
Figure 1: Effect of total future steps on evaluation.



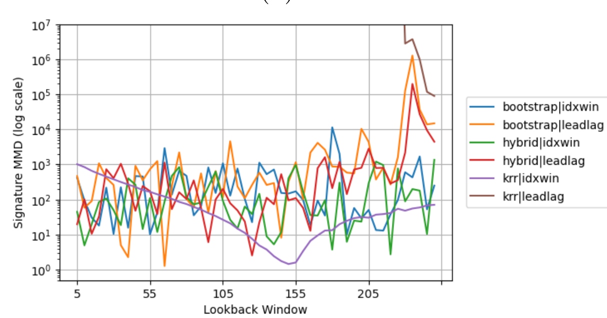
(a) GBM



(b) CEV



(c) Merton Jump Diffusion



(d) Variance Gamma

Figure 2: Effect of lookback window on evaluation.